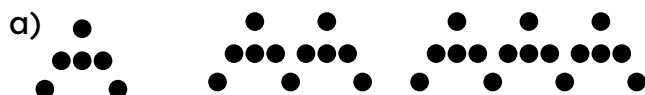




Generating a sequence using a position-to-term rule

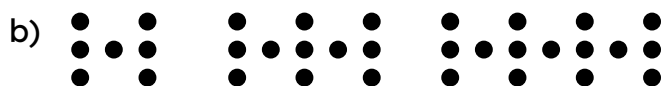
Task A: Writing position-to-term rules

- 1) For each sequence, write a **position-to-term** rule for the number of dots in each pattern.



Term number	1	2	3
Term	6	11	16

Rule: 'term number $\times 5 + 1$ '



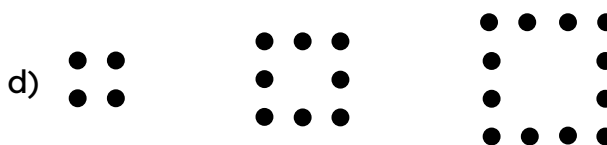
Term number	1	2	3
Term	7	11	15

Rule: 'term number $\times 4 + 3$ '



Term number	1	2	3
Term	7	13	19

Rule: 'term number $\times 6 + 1$ '



Term number	1	2	3
Term	4	8	12

Rule: 'term number $\times 4$ '

- 2) Four students are writing a **position-to-term** rule for the sequence:

9, 11, 13, ... Who is correct?

For the incorrect answers, show how you know they are incorrect.

The term is the term number multiplied by 9



Sam

The term is the term number add 2



Sofia

The term is the term number multiplied by 2 add 7



Alex

The term is the term number multiplied by 2 add 9



Izzy

Sam: $1 \times 9 = 9$

But 2×9 is not 11

Sofia: The term-to-term rule could be add 2.

But $1 + 2 \neq 9$, $2 + 2 \neq 11$...

The term number add 2 is not the correct rule.

Alex is correct.

$$T_1 = 2 + 7,$$

$$T_2 = 2 + 2 + 7,$$

$$T_3 = 2 + 2 + 2 + 7$$

Izzy: $1 \times 2 + 9 \neq 9$

and $2 \times 2 + 9 \neq 11$



Task B: Generating terms from position-to-term rules

For each sequence, find the value of T_{60}

- | | |
|---|----------------|
| a) Each term is the term number multiplied by 5 plus 6 | $T_{60} = 306$ |
| b) Each term is the term number halved then add 2 | $T_{60} = 32$ |
| c) Each term is the term number subtract 5 | $T_{60} = 55$ |
| d) Each term is 10 less than the term number then multiplied by 3 | $T_{60} = 150$ |
| e) 12, 15, 18, 21, ... (with a constant additive pattern) | $T_{60} = 189$ |
| f) 18, 19, 20, 21, ... (with a constant additive pattern) | $T_{60} = 77$ |
| g) 1.4, 1.8, 2.2, 2.6, ... (with a constant additive pattern) | $T_{60} = 25$ |

